

Development Of The Dental Pulp Root Formation

Development Of The Dental Pulp

Starts at the **bud stage** as condensation of mesenchymal cells

Some of these cells will become odontoblasts , those come from the neural crest (ectomesenchyme)

The others are of mesenchymal origin

All cells look the same under the microscope

These cells rapidly divide and are separated by little extracellular fluid.

Development Of The Dental Pulp

In the **cap stage** the cells forming this mass are known as the **dental papilla**

Dental follicle : mesenchymal cells surrounding the developing enamel organ and later on they will form the periodontal ligament and the supporting tooth structures.

During the bell stage cyto-differentiation of cells by signals from internal enamel epithelium take place into :

1. Central mass of fibroblasts
2. Peripheral layer of odontoblasts.

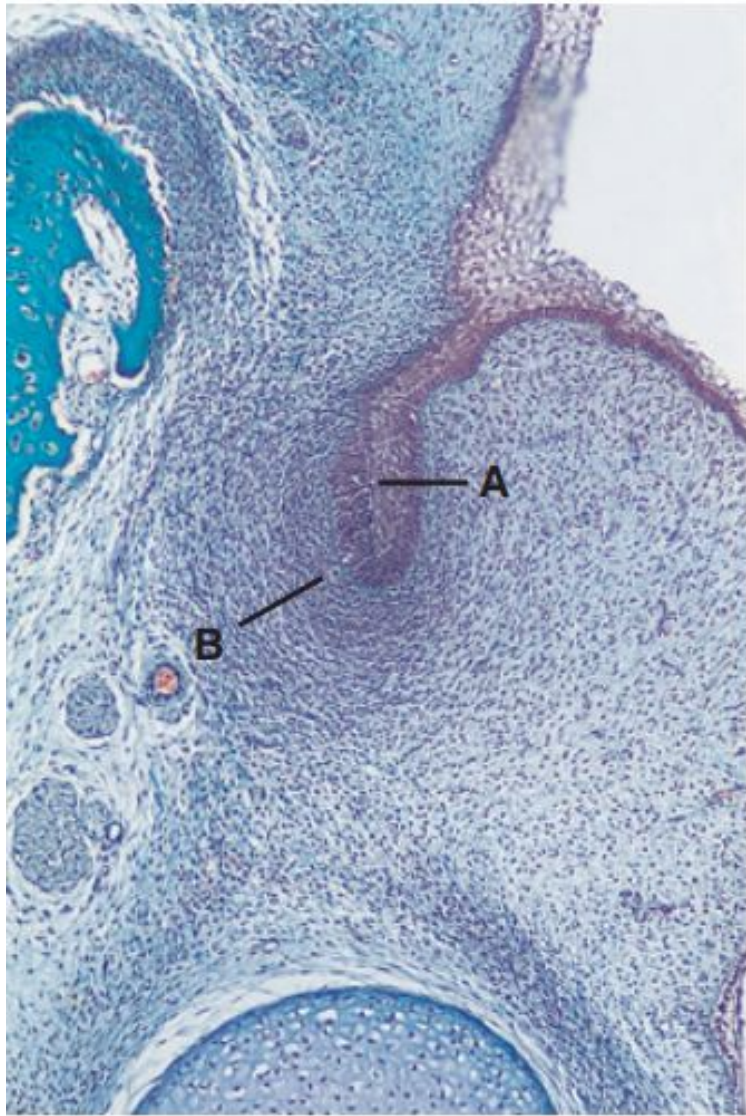


Fig. 24.1 A developing enamel organ at the early bud stage (A), with a condensation of mesenchymal cells (B) around it (Masson's trichrome stain; $\times 80$).



Fig. 24.2 An enamel organ at the cap stage of development (A) surrounding the mesenchymal cells of the dental papilla (B). The tooth germ is surrounded by the developing dental follicle (C) (H & E; $\times 80$). Courtesy Prof. M M Smith.

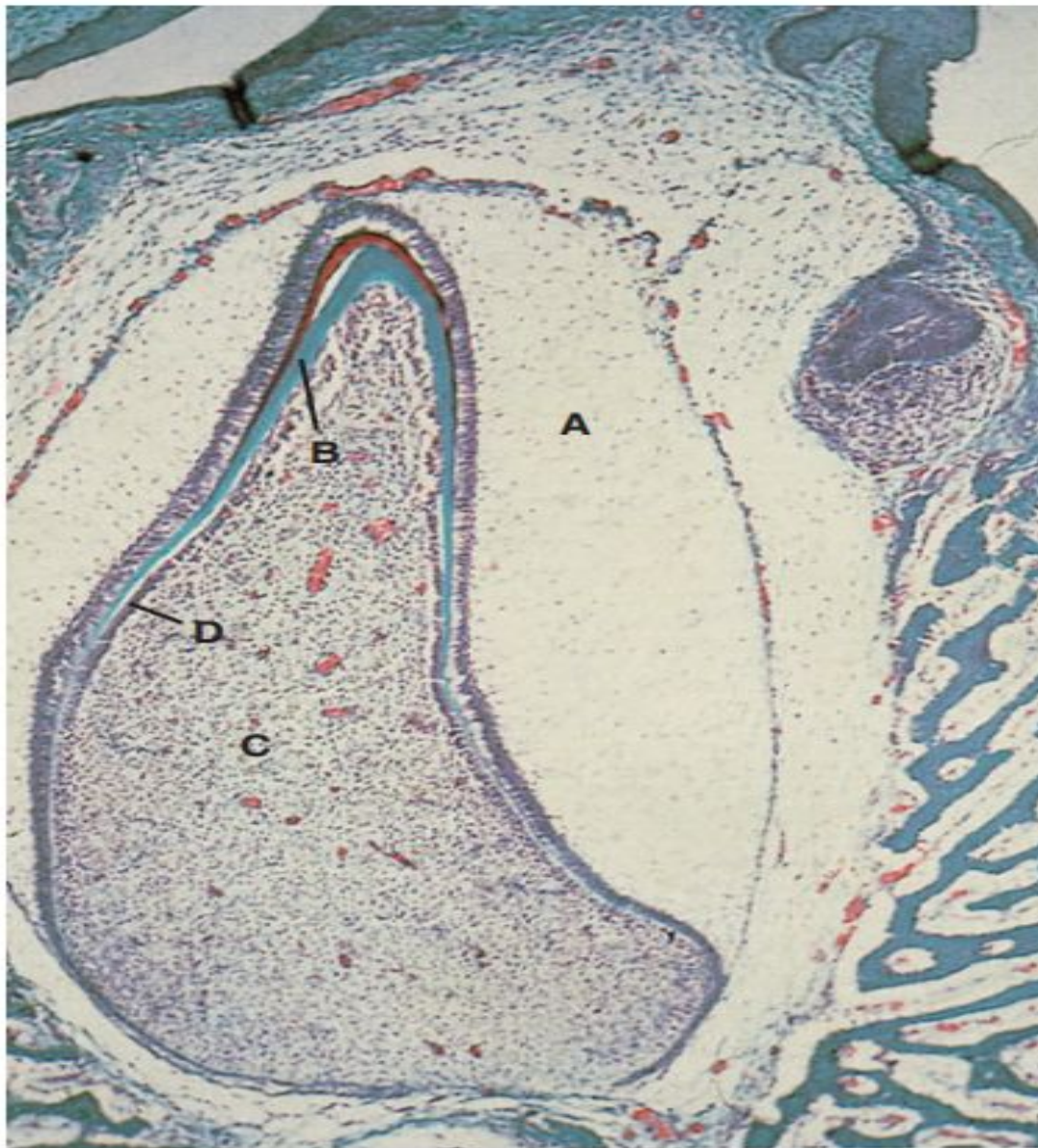


Fig. 24.3 A developing tooth at the late bell stage of development (appositional phase). A = enamel organ; B = developing dentine; C = dental papilla; D = odontoblast layer (Masson's trichrome; $\times 55$).

Development Of The Dental Pulp

The **dental papilla** becomes the **dental pulp** once the first layer of dentin has been laid.

The remaining ectomesenchymal cells are undifferentiated , they become packed together and they have large nucleus and very little cytoplasm and are responsible for formation of coarse fiber bundles which appear at tooth maturation.

Cells differentiation in **hertwig root sheath** determines the final morphology of pulp space .

The development of pulp is considered complete once the final length of the roots has been reached.

Cell rich zone : layer present under the odontoblasts , formed by migration of central cells.

Cell free zone : some believe it is a fixation artifact , appears around tooth eruption.

Cells of dental pulp can differentiate to different types to respond to insult or stimulus or whatever.

Blood Supply

During the bell stage branches of the jaws blood supply enter the base of the papilla.

Some of these branches enlarge and become the basic arterioles of the dental papilla

They form a complete network of arterioles and veinules in the odontoblast and the sub odontoblast layer.

The vascularity of odontoblast increase with increasing dentin thickness as a result of cells invasion into the vascular bed.

Exact lymphatic pattern and timing of pulp is not clear yet.

Mature pulp cells that enter with the blood vessels are : pericytes , lymphoid cells and macrophages.

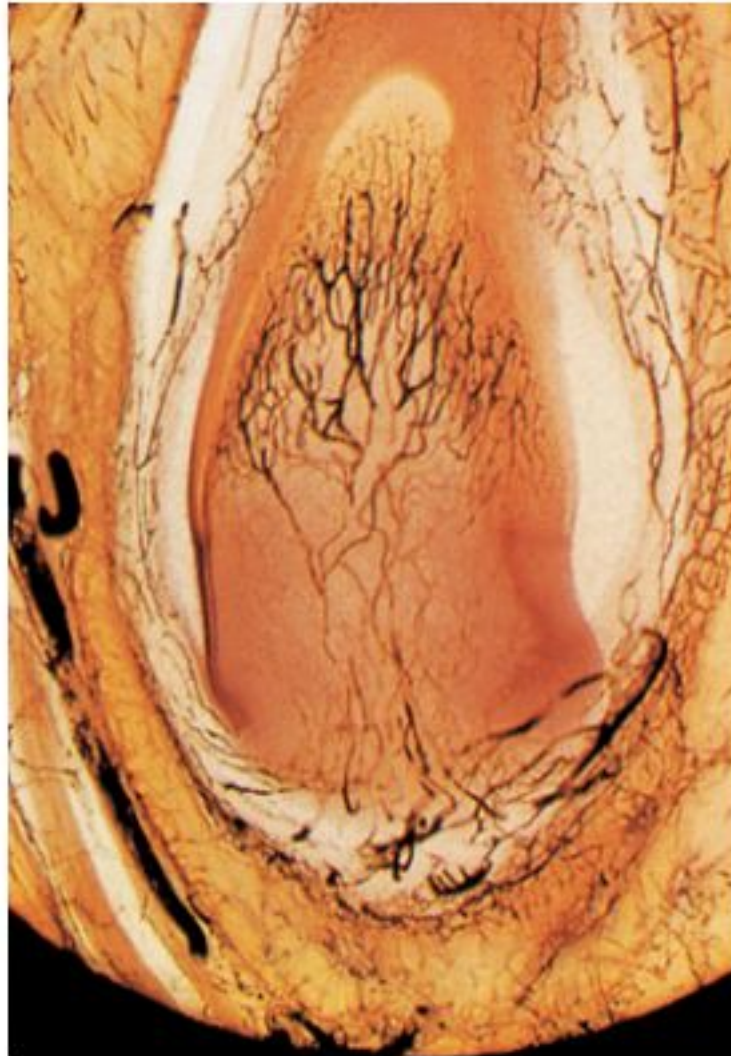


Fig. 24.5 The vascular pattern within the dental papilla at the late bell stage of development. The vascular system has been perfused with Indian ink (H & E; $\times 40$). Courtesy Dr D Adams.

Nerve Supply

Nerve fibers enter the dental papilla at a late stage.

First fibers to enter the papilla are from the trigeminal nerve close to the blood vessels

Those fibers are **sensory autonomic fibers** and they control blood flow, this might help them to control tooth development as well.

Neurotrophin brain derived growth factor is seen at the periphery of developing pulp and this indicates

That this region is innervated by trigeminal nerve.

Nerve fibers enter the pulp before final root formation but the formation of plexus such as plexus of Raschkow takes place after root formation is complete.

Root Development

It starts with axial tooth eruption.

Interaction between the dental follicle , the dental papilla and Hertwig epithelial root sheath (derived from cervical loop region of enamel) .

At the late bell stage root formation starts when dentinogenesis and amelogenesis are advanced , internal and external epithelial layers form and proliferate apically to determine root formation.

Number of roots is determined by number of epithelial shelves (these also proliferate to form multiple secondary apical foramina).



Fig. 25.6 The developing root. A = epithelial root sheath; B = dental papilla; C = primary apical foramen; D = dental follicle; E = developing root dentine; F = odontoblast layer (H & E; $\times 32$).

Root Development

Root eruption takes place when two thirds of root is complete , in this case we have open wide apex apically.

And it closes 3 years later.

During eruption epithelial root sheath encloses the enamel organ except for the apical foramen.

Between the 2 epithelial layers surrounding the dental papilla there is no stratum intermedium or stellate reticulum , these are seen only in region of enamel pearls..

Dental follicle is external to root sheath and forms the cementum alveolar bone and periodontal ligament.

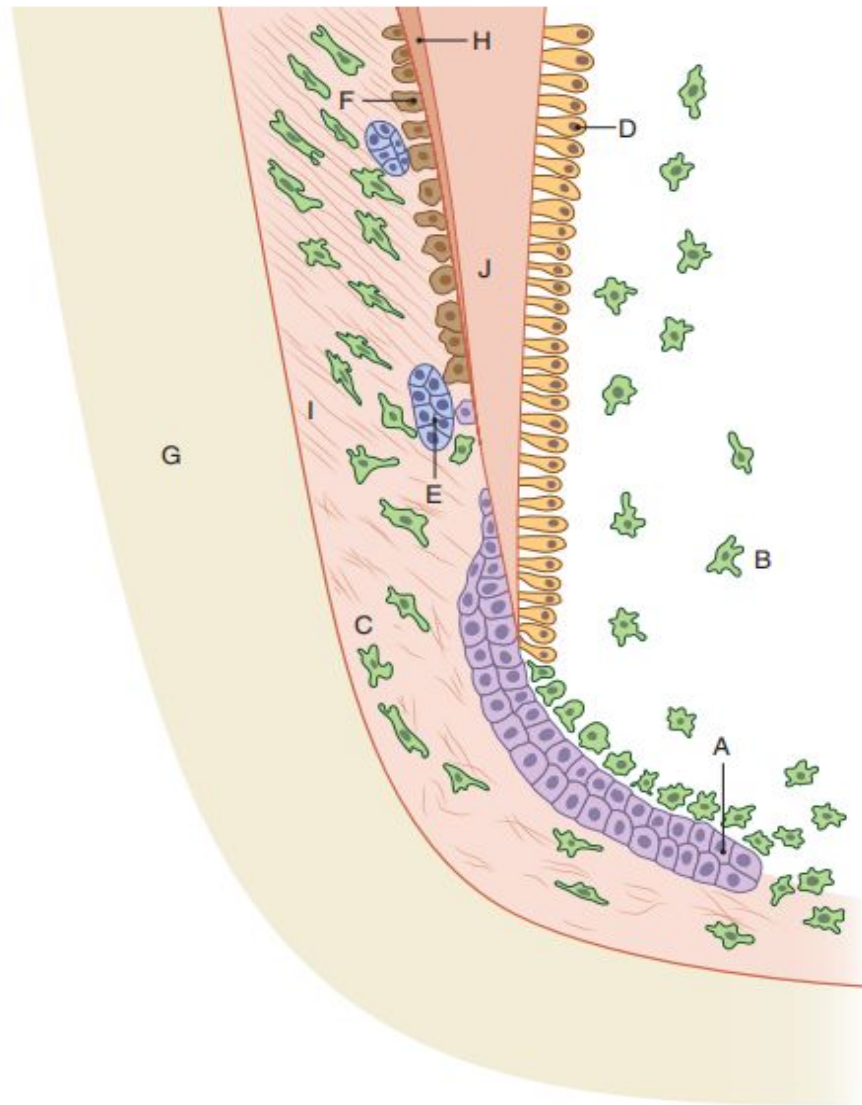


Fig. 25.7 Schematic drawing of the developing root. A = epithelial root sheath; B = dental papilla; C = dental follicle; D = odontoblasts; E = epithelial rests; F = cementoblasts; G = developing alveolar bone; H = developing cementum; I = developing periodontal ligament; J = root dentine.

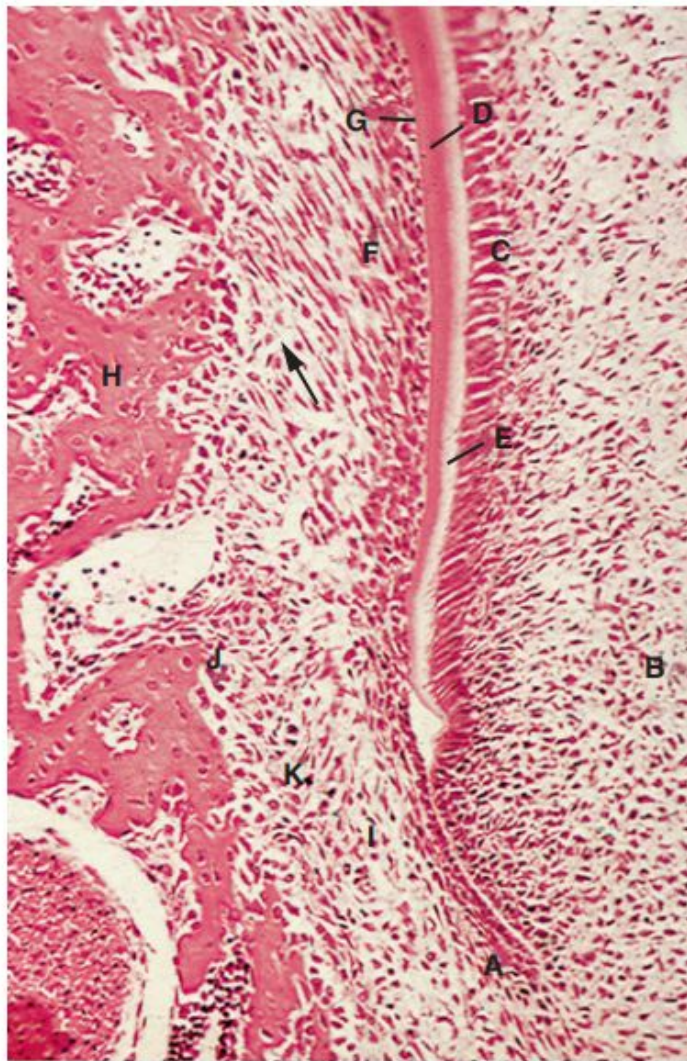


Fig. 25.8 The apical region of the developing root, periodontal ligament and alveolus. A = epithelial root sheath; B = dental papilla; C = odontoblasts; D = dentine of root; E = predentine layer; F = cementoblasts; G = developing cementum; H = developing alveolar bone; I = inner investing layer of dental follicle; J = outer layer of the dental follicle; K = intermediate layer of dental follicle. Arrow indicates oblique orientation of follicular tissue with formation of cementum (H & E; $\times 150$).

Once cementogenesis has begun , cells of dental follicle become obliquely oriented to root surface and their organelles increase and become the fibroblasts of PL , and they secrete collagen in the extracellular fluid and become embedded in cementum on the root surface and in bone on alveolar surface.

Root Development

The cells of the internal layer of root epithelium stimulate the cells of the dental papilla to become odontoblasts.

After odontogenesis start , epithelial root sheath cells loose their continuity and become **rest cells of malassez**.

And the mesenchymal cells of the dental follicle start differentiating into cementoblasts and cementogenesis start.

Cushion Hammock Ligament

Connective tissue lying immediately below the developing root apex was described as fibrous network with fluid filled compartments , connected to the alveolus on either side

Was thought to provide a strong base to prevent bone resorption by heavy forces of eruption underneath the developing root, this view is no longer held.

It was proven that this fibrous membrane is not attached to bone but merges with fibers of PL , so more correctly was called the **pulp limiting membrane**

Its surgical removal does not affect tooth eruption.



Fig. 25.9 The pulp-limiting membrane (arrowed) (Mallory's trichrome; $\times 30$). Courtesy Dr D Adams.

It was proposed that changes in vascular permeability around developing roots can affect eruption behavior.

Dense accumulation of fluid (effusion) was seen beneath root apices , when they used radioactive fibrinogen to study this , it was quickly incorporated into the effusion .

Vascular hypotheses of eruption

Effusion forces the root and the bone apart on eruption , and this contributes to eruption and further root growth.

Root Development

Formation of collagen fibers within the periodontal ligament

- ❖ **stage 1:** before eruption the dentogingival and oblique fibers are formed in permanent molars. In premolars only the dentogingival are formed.
- ❖ **stage 2:** as the tooth erupts in the oral cavity , PL of permanent molars is well organized. In premolars PL fibers are organized coronally only (near the alveolus) apically they are present yet they are not organized.
- ❖ **stage 3:** in molars cervical fibers become organized in occlusion. In premolars cervical fibers become organized and apically are not organized.
- ❖ **stage 4:** after a while in function , fibers of all teeth become organized.

Formation Of Collagen Fibers Within The Periodontal Ligament

Bulk of PL fibers is not fully organized during eruption and this was proposed to help dissipate the forces during tooth eruption.

The more time in function , the inclination of fibers decrease and the thickness increase , however this differ between species.

During tooth eruption , active resorption takes place at the base of the socket. , Some species have the opposite e.g. The dogs premolars ! this depends on the distance of eruption , if the distance more than the root length bone deposition is a must.

Cementogenesis

Defined as the formation of primary (acellular cementum) and secondary cellular cementum.

Primary Acellular Cementum

Cementogenesis begins at the cervical margin and progress downward.

Epithelial root sheath cells do not enlarge in this phase in contrast to enamel organ cells during amelogenesis.

The epithelial root sheath is separated from both the dental follicle and the dental papilla by **basal lamina**.

Interaction between epithelial root sheath and cells of dental papilla induces them to differentiate into **odontoblasts**.

First formed layer of dentine is called **predentin** : structureless organic glass like matrix under the polarized light. called [hyaline layer of Hopewell-Smith](#) once it is fully mineralized and is about 10 microns.

Cells of the epithelial root sheath are close to the predentine cells for a short distance only , this allows the fibroblast like cells of dental follicle -represent cementoblasts to be close and interact with the unmineralized hyaline layer.

At their deep surface they unite with the hyaline layer cells , and on their superficial surface they form **fibrous fringe** extending perpendicular in PL space for 10-20 microns.

At the beginning of **cementogenesis** epithelial root sheath cells secrete proteins such as ameloblastin +/- amelogenin into the united interface because they have Golgi Complex in their cells . Their function is unclear; might relate to epithelial mesenchymal interactions to induce cementoblasts or odontoblasts.

During subsequent mineralization they are lost , may remain some of them in dentin granular layer

As the odontoblasts move to the pulp , they leave their process behind forming **the Granular Layer of Tomes** and then **Circumpulal** dentine (basic tubular structure of dentin).

Mineralization starts few microns from the hyaline layer, thus the innermost layer of it is continuous with the periphery of the pulp .

Similar staining pattern is observed after administration of antibiotic tetracycline.

Mineralization of cellular cementum spreads outward , and this layer is firmly attached to root dentin.

At this stage, PL fibers are more parallel to root surface and not attached to root fringe.

In early stages of cementogenesis , cementoblasts secrete many proteins such as bone sialoprotein and osteopontin. And they play a role in cell attraction and differentiation. They might be related to a type of osteopetrosis in which osteocytes are deficient. Subsequent development of acellular cementum includes

1. Slow increase in thickness,
2. Continuity between PL fibers and fibrous fringe at the surface of root dentine-- this is the only way a tooth can be supported in the socket and timing is different between permanent and primary teeth. For permanent teeth this attachment happens once the tooth erupts and acellular cementum thickness is 10 micron.

Acellular cementum lining the tooth before this time is known as intrinsic acellular cementum

3. Slow mineralization of collagen.

Once complete union has occurred between PL fibers and surface of cementum:

This cementum is classified as **Acellular extrinsic fiber cementum** , increases in thickness about 2-2.5 microns throughout life , the cells cannot be readily recognized as a layer from the surrounding , they rather mingle .

Cementoblasts contain organelles responsible for protein synthesis , and they secrete the surrounding ground material around the collagen bundles.

Mineralization of cementum is initiated by the hydroxyapatite crystals in the adjacent dentin not by cells. P1 fibroblasts also play a role in mineralization.

Mineralization is a long slow process , this explains the absence of a pre cementum layer.

Mineralization of cementum is linked to the initial root dentine formation , thats why drugs which inhibit this process such as bisphosphonates inhibit also cementogenesis.

Cementogenesis is rythematic process with periods of increased and decreased activity , this explains the presence of **incremental lines** , incremental lines are associated with decreased activity , they contain high content of ground material and water and less amount of collagen.

Acellular cementum forms more slowly , than cellular cementum , the incremental lines are packed more together.

Acellular afibrillar cementum

Thin layer overlying enamel in cervical tooth area.

Reduced enamel epithelium overlying tooth becomes damaged , cells of connective tissues adjacent to enamel organ differentiate into cementoblasts which secrete afibrillar matrix and then calcifies .

Secondary Cellular Cementum

Forms in the apical region and furcation at time of tooth eruption.

Formation is similar to primary cementum and is related to dentin formation.

Following loss of continuity of epithelial root sheath , large basophiles of dental follicle differentiate into cementoblasts near root surface.

Under **light microscope** , they have endoplasmic reticulum and secrete the matrix and the collagen fibers of the **secondary cellular cementum**.

Fibers are parallel to root surface , with increased rate formation , unmineralized layer of pre cementum forms around 5 microns thickness.

Overall secondary cementum is less mineralized than primary cementum , mineralization takes place in linear matter.

Cells become gradually incorporated within the matrix they secrete , then the become **osteocyte** ,

Stem cells from the periodontal ligament (PL) or dental follicle can differentiate into **cementoblasts**, which form cementum, the bone-like tissue covering the tooth root.

Cellular cementum has no Sharpey's fibers and is present only in apical and furcation areas , it alternates with acellular extrinsic fiber cementum to form **mixed stratified cementum**.

When cellular intrinsic fiber cementum is covered by acellular extrinsic fiber cementum, Sharpey's fibers are attached to root surface , in the opposite situation , Sharpey's fibers are detached from root surface.

Cellular Mixed Fiber Cementum

Cellular intrinsic cementum gives attachment to extrinsic fibers from PL which are aligned perpendicular to root surface in contrast to intrinsic fibers secreted by cementoblasts themselves which are secreted parallel to root surface.

As for acellular cementum , exact origin of cells responsible for formation of secondary cementum still needs clarification , it is believed they come from different cell populations.

Enamel pearls

Small spherical of enamel located on tooth surface near cervical end popular at the bifurcation,

In region affected : stellate reticulum and stratum intermedium develop between external and internal enamel sheath



Fig. 25.36 An enamel pearl located near the root bifurcation area (arrow). Courtesy Dr C Franklyn.